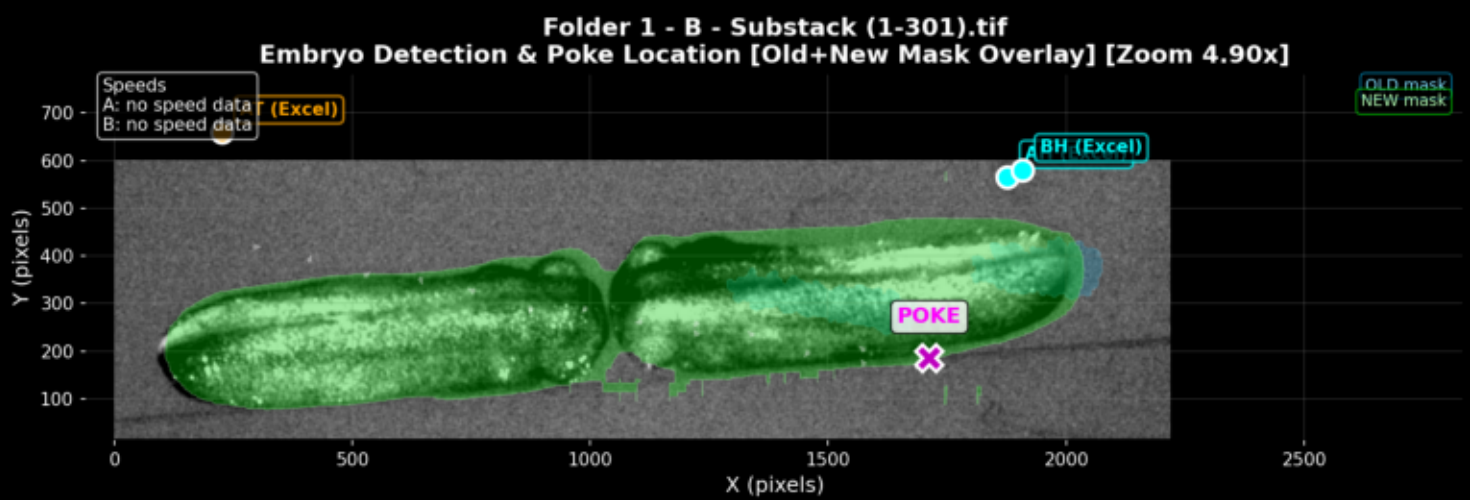
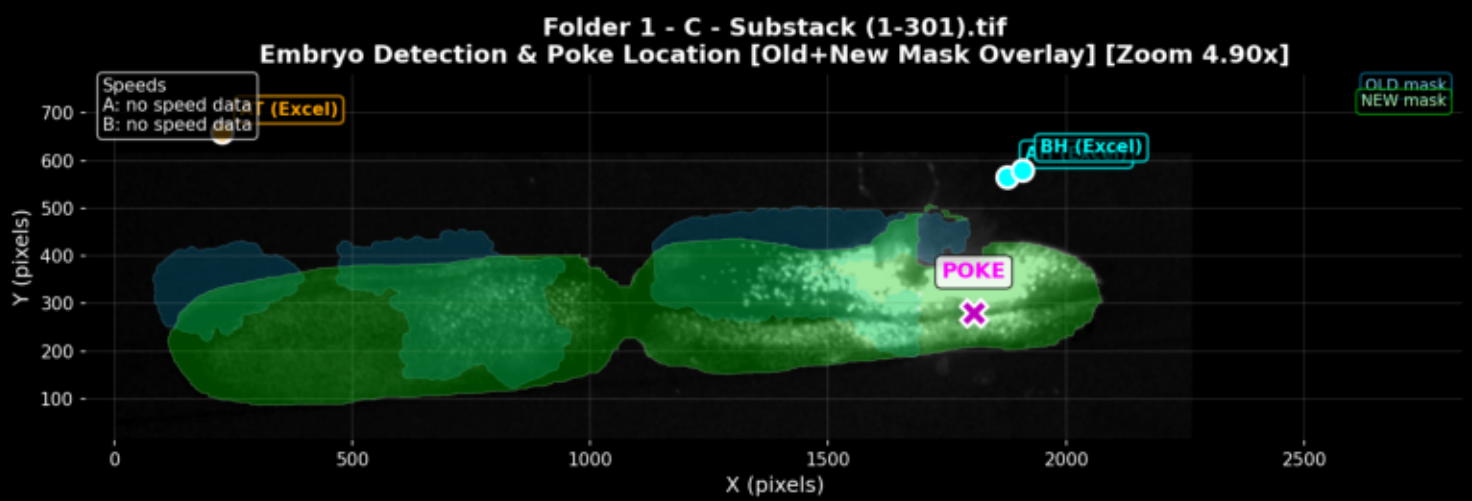
Embryo Detection Summary

Folder	Video	Embryo A	Embryo B	Poke Location	Healed Wound
1	B - Substack (1-301).tif	not detected	not detected	(1713.0, 185.0)	not detected
1	C - Substack (1-301).tif	not detected	not detected	(1806.2, 279.5)	not detected
12	B - Substack (1-301).tif	not detected	head left, tail right	(1395.8, 440.2)	not detected
12	C - Substack (45-421).tif	not detected	head left, tail right	(1390.8, 413.7)	not detected
16	B 10 min after wound - Substack (1-301).tif	head right, tail left	head right, tail left	(1130.5, 292.0)	not detected
16	B 5 min after wound - Substack (1-301).tif	head right, tail left	head right, tail left	(1165.1, 285.8)	not detected
16	C after wound - Substack (55-601).tif	head right, tail left	head right, tail left	(1167.5, 342.0)	not detected

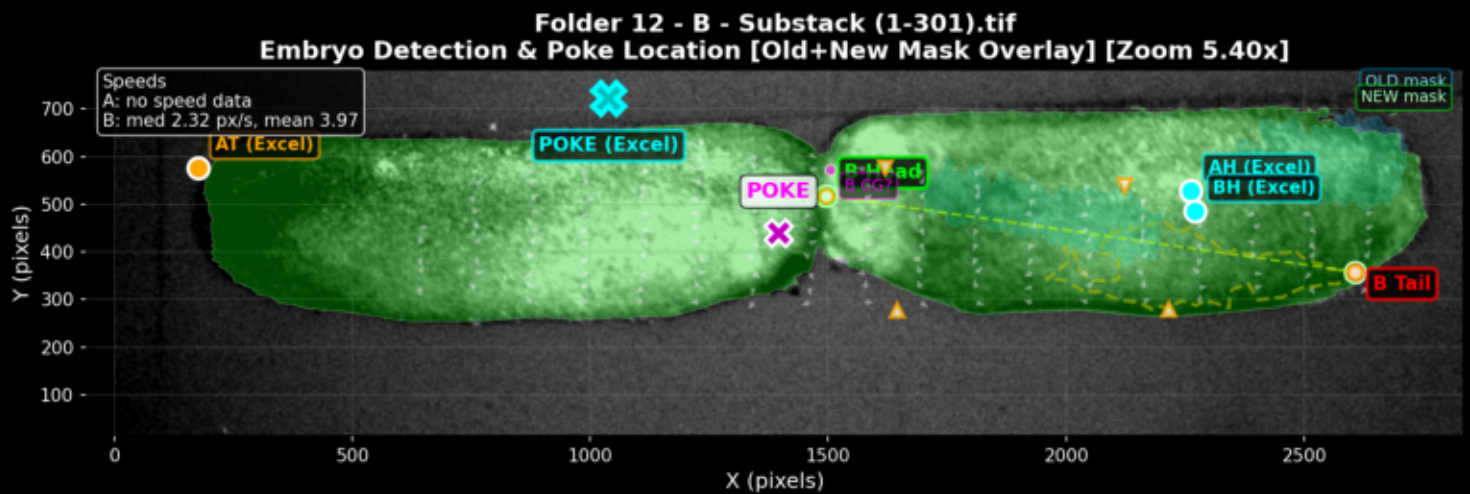
Folder 1a: B - Substack (1-301).tif



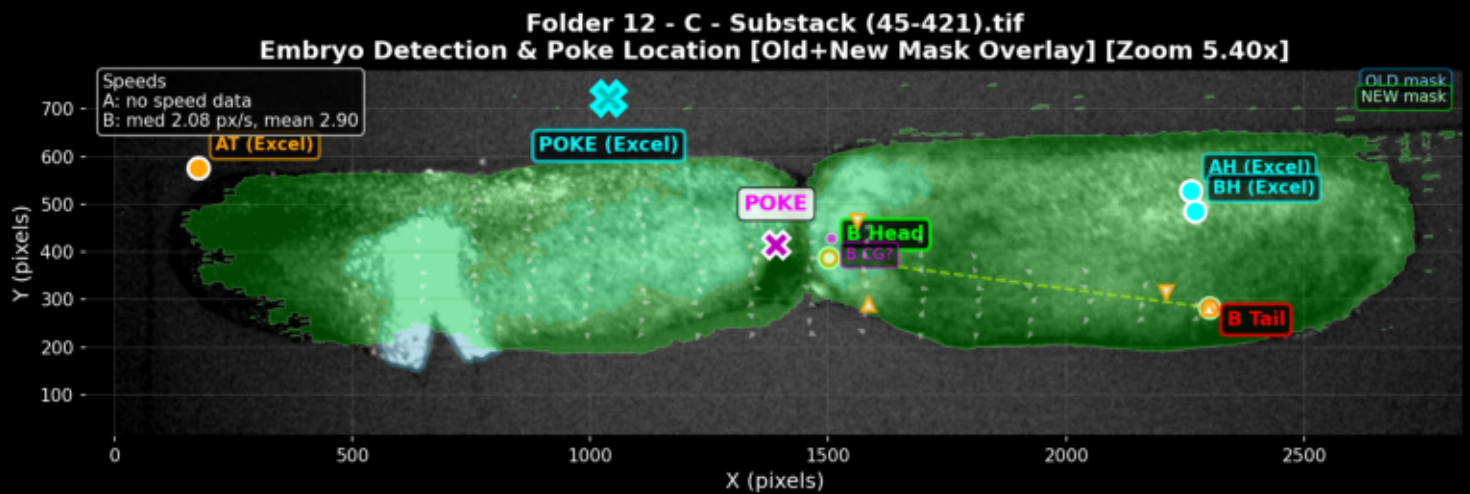
Folder 1b: C - Substack (1-301).tif



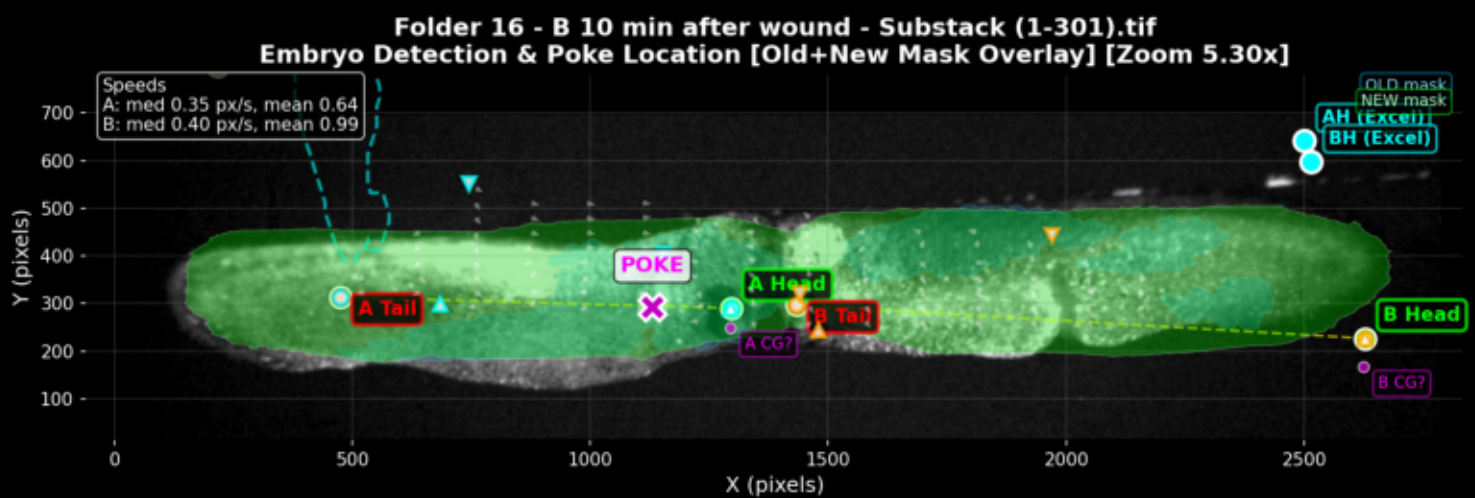
Folder 12a: B - Substack (1-301).tif



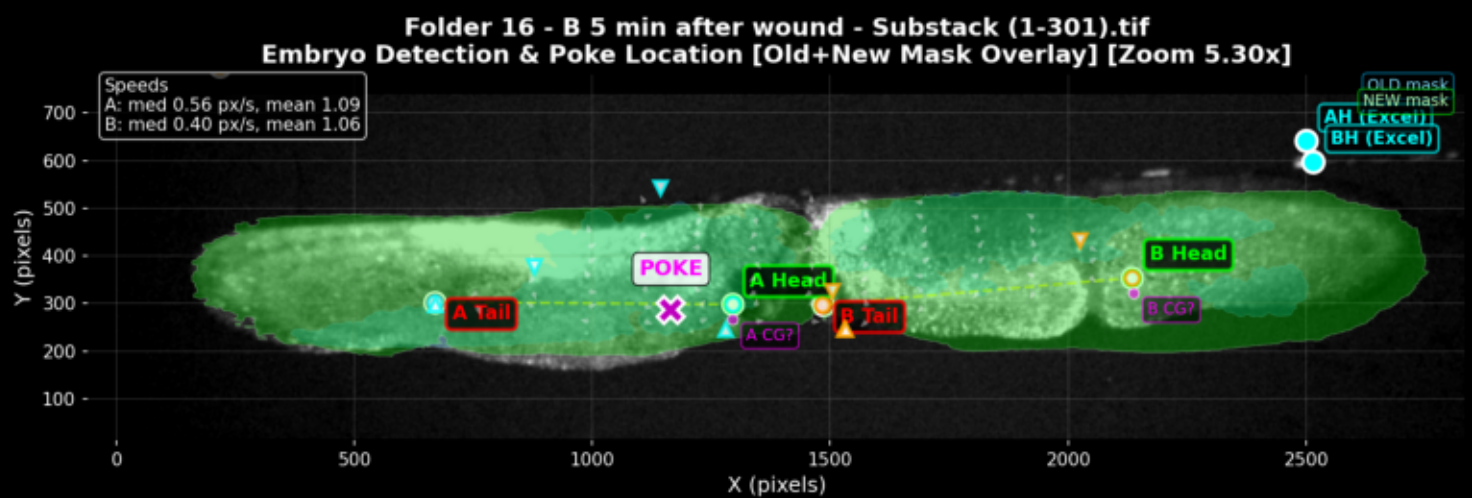
Folder 12b: C - Substack (45-421).tif



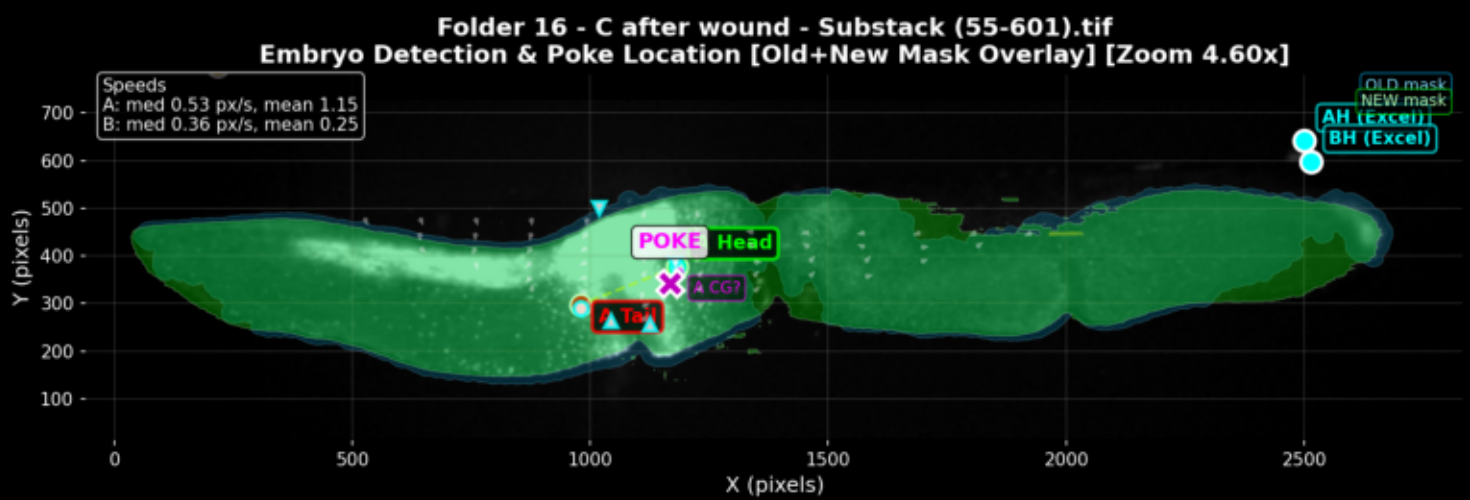
Folder 16a: B 10 min after wound - Substack (1-301).tif



Folder 16b: B 5 min after wound - Substack (1-301).tif



Folder 16d: C after wound - Substack (55-601).tif



Detection Warnings and Validation Log

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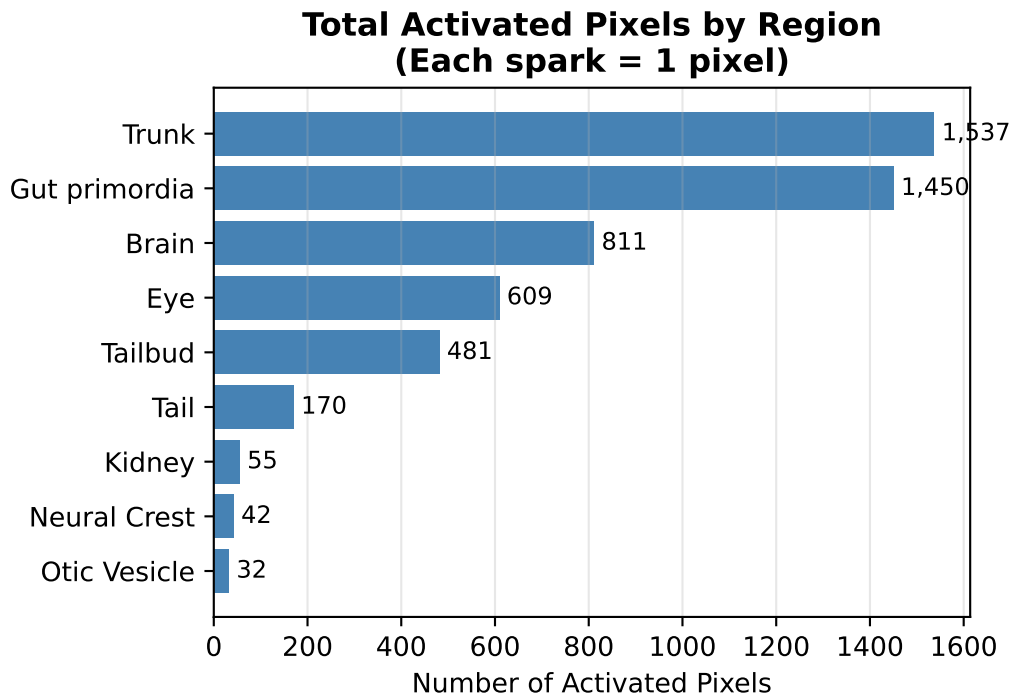
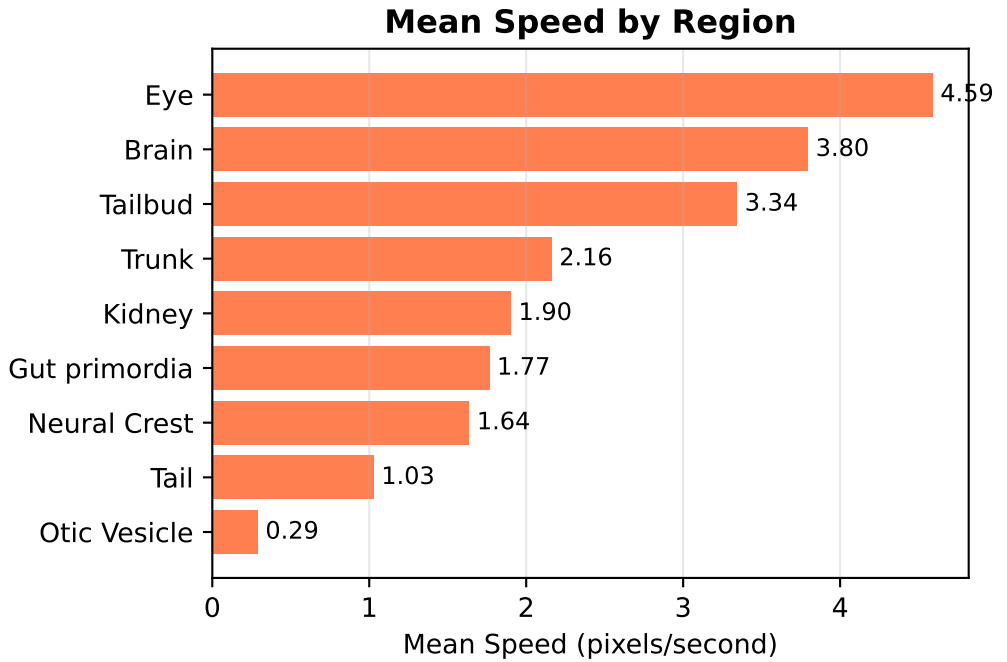
Embryo Detection Warnings and Validation Log
================================================================================

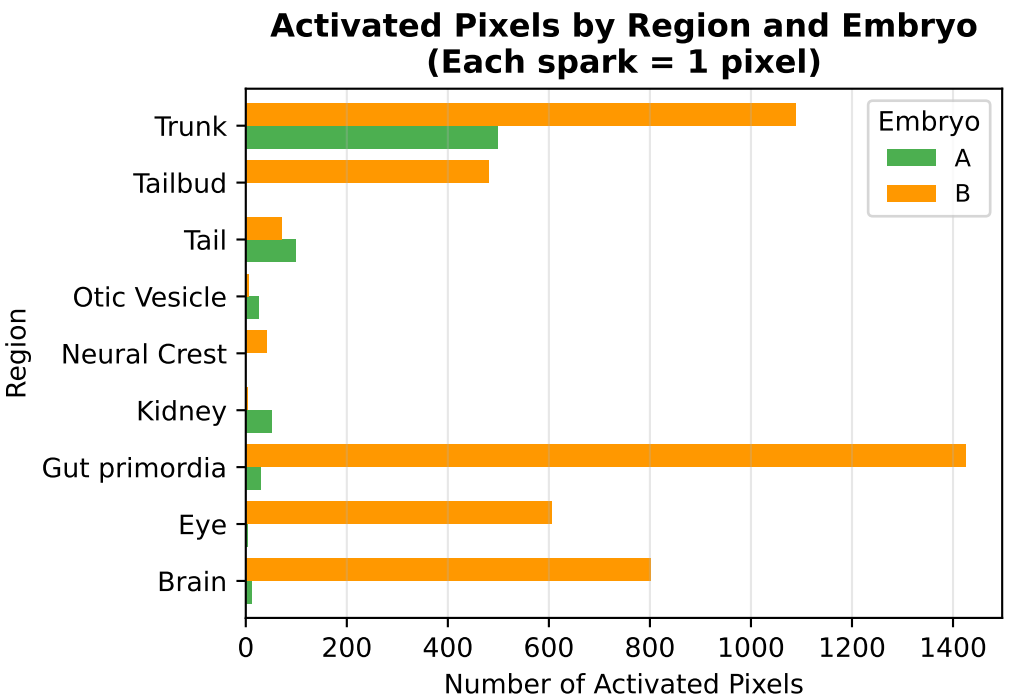
All warnings and validation messages from embryo detection:

    [Embryo A] ⚠ CRITICAL: Tail at x=2219.0 is on RIGHT side (center=1110.0)
    [Embryo A] ⚠ CRITICAL: Violates spatial constraints
    [Embryo B] ⚠ CRITICAL: Head at x=610.0 is on LEFT side (center=1132.5)
    [Embryo B] ⚠ CRITICAL: Tail at x=1042.0 is on LEFT side (center=1132.5)
    [Embryo A] ⚠ CRITICAL: Violates spatial constraints
    [Embryo A] ⚠ CRITICAL: Tail at x=2758.0 is on RIGHT side (center=1452.0)
    [Embryo A] ⚠ CRITICAL: Violates spatial constraints
    [Embryo B] ⚠ CRITICAL: Violates spatial constraints
    [Embryo A] ⚠ CRITICAL: Tail at x=2727.0 is on RIGHT side (center=1465.0)
    [Embryo A] ⚠ CRITICAL: Violates spatial constraints
    [Embryo B] ⚠ CRITICAL: Violates spatial constraints
    [Embryo A] ⚠ CRITICAL: Tail at x=2640.0 is on RIGHT side (center=1394.0)
    [Embryo A] ⚠ CRITICAL: Violates spatial constraints
    [Embryo B] ⚠ CRITICAL: Violates spatial constraints
    [Embryo A] ⚠ CRITICAL: Tail at x=2703.0 is on RIGHT side (center=1437.0)
    [Embryo A] ⚠ CRITICAL: Violates spatial constraints
    [Embryo B] ⚠ CRITICAL: Violates spatial constraints
    ⚠ Skipping single-frame TIFF: C after wound - MAX_colored.tif (1 frame(s))
    [Embryo A] ⚠ CRITICAL: Tail at x=2685.0 is on RIGHT side (center=1417.5)
    [Embryo A] ⚠ CRITICAL: Violates spatial constraints
    [Folder 16, Video C after wound - Substack (55-601).tif] [Embryo B] ⚠ CRITICAL: Head-tail distance too short (65.7px < 144.8px)
    ⚠ Skipping single-frame TIFF: TMR.tif (1 frame(s))
```

Sparks Summary by Anatomical Region (Pixel-Based)





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SUMMARY STATISTICS (Pixel-Based)

Total Activated Pixels: 15,117
Pixels with Region: 4,741 (31.4%)
Total Detection Events: 57,493
Unique Regions: 9

TOP 5 REGIONS BY PIXEL COUNT:
1. Tailbud: 481 pixels (10.1%)
2. Eye: 609 pixels (12.8%)
3. Brain: 811 pixels (17.1%)
4. Gut primordia: 1,450 pixels (30.6%)
5. Trunk: 1,537 pixels (32.4%)


SPEED STATISTICS:
Overall Mean Speed: 2.25 px/s
Overall Median Speed: 0.81 px/s
Fastest Region: Eye (4.59 px/s)
```